12 Expand and simplify $(x - 2)(3x + 2)(2x + 3)$

(Total for Question 12 is 3 marks)

13 In a school there are 16 teachers and 220 students.
Of these students 120 are girls and 100 are boys.

One teacher, one girl and one boy are going to be chosen to represent the school.

Work out the number of different ways there are to choose one teacher, one girl and one boy.

16 choices of teacher, 120 of girl, 100 of boy.

Product rule: 16 x 120 x 100 = 192000.

1 mark
multiply independent choices

1 mark
192000

192000

(Total for Question 13 is 2 marks)

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